

For Loop (10 Problems)

1. Print numbers from 1 to 10

```
#include <iostream>
using namespace std;

int main(){
    int num = 10;
    for(int i = 1; i <= 10; i++){
        cout << i << " ";
    }
}
```

2. Print numbers from 10 to 1

```
#include <iostream>
using namespace std;

int main(){
    // int num = 10;
    // for (int i = num; num--; i--){
    //     cout << i << " ";
    // }

    int num;
    cout << "Enter a Number: ";
    cin >> num;
    // for(int i = 1; num >= i; num--){
    //     cout << num << " ";
    // }
    for (num; num >= 1; num--){
        cout << num << " ";
    }
}
```

3. Print the multiplication table of a number

```
#include <iostream>
using namespace std;

int main(){
    int num;
    cout << "Enter Number: ";
    cin >> num;
```

```
for(int i = 1; i <= 10;i++){
    cout <<num <<" * " <<i <<" = " <<num * i <<endl;
}

return 0;
}
```

4.Find the sum of numbers from 1 to N

```
#include <iostream>
using namespace std;
// Find the sum of numbers from 1 to N.
int main(){
    int num;
    cout << "Enter a Number: ";
    cin >> num;

    int sum = 0;

    for (int i = 1;i<=num;i++){
        sum = sum + i;
        if(i <= num){
            if(i==num){
                cout <<i << " = ";
            }else{
                cout <<i <<" + ";
            }
        }
    }
    cout << sum;

    return 0;
}
```

5.Find the factorial of a number

```
#include<iostream>
using namespace std;

// Find the factorial of a number.
int main(){
    int num;
    cout <<"Enter a Number: ";
    cin >> num;
```

```
double fact = 1;
for(int i = 1; i <= num ;i++){
    fact = fact * i;
    if(i <= num){
        if(i==num){
            cout <<i <<" = ";
        }else{
            cout <<i <<" * ";
        }
    }
}
cout <<fact ;

return 0;
}
```

6 . Print all even numbers from 1 to 100

```
#include <iostream>
using namespace std;

int main(){
    int num;
    cout <<"Enter a number: ";
    cin >> num;

    cout <<"All Even Numbers:" <<endl;
    for(int i = 1;i <=num ;i++){
        if( i % 2 == 0){
            cout << i <<" ";
        }
    }

    return 0;
}
```

7.Print all odd numbers from 1 to 100

```
#include <iostream>
using namespace std;

int main(){
    int num;
    cout <<"Enter a number: ";
    cin >> num;

    cout <<"All ODD Numbers:" <<endl;
    for(int i = 1;i <=num ;i++){
```

```
        if( i % 2 != 0){
            cout << i <<" ";
        }
    }

    return 0;
}
```

8.Print squares of numbers from 1 to N

```
#include <iostream>
using namespace std;
// Print squares of numbers from 1 to N.
int main(){
    int num;
    cout <<"Enter a number: ";
    cin >>num;

    int squ;

    for(int i = 1;i <= num;i++){
        squ = i * i ;
        cout <<squ <<" ";
    }

    return 0;
}
```

9.Count numbers divisible by 3 from 1 to N

```
#include <iostream>
using namespace std;
#define g "\033[32m"
#define y "\033[33m"
#define reset "\033[0m"

int main(){
    int num;
    cout <<"Enter a Number: ";
    cin >>num;

    int count = 0;
    for(int i = 1;i <=num;i++){
        if(i % 3 == 0){
            cout <<i <<" ";
        }
    }
}
```

```

        count++;
    }
}
cout <<endl;
cout <<"Count of 1 to " <<num <<" = " << count <<" numbers divisible by 3." ;

return 0;
}

```

10 . Print this pattern

```

*
**
***
****
*****

```

```

#include <iostream>
using namespace std;

int main(){
    int num_str;
    cout <<"Number Of Star: ";
    cin >> num_str;

    for (int i = 1;i <=num_str ;i++){
        for (int j = 1;j <= i;j++){
            cout <<" * " ;
        }
        cout <<endl;
    }
}

```